

Parallel depth batch 4 — Cosmological Λ CDM + strong-interaction + atomic constants (all CANDIDATE-ARITHMETIC tier per Keeper v1.4 brake). New candidates: $\alpha_s(m_Z) \approx \text{rank}^4/N_{\text{max}} = 16/137$ (0.94% off); $\Omega_m \approx \text{rank} \cdot N_c \cdot g / N_{\text{max}} = 42/137$ (2.7%); existing $n_s = 1 - n_C/N_{\text{max}}$ (0.14%). Atomic constants (Rydberg, Bohr) derive from $m_e + \alpha$. ALL CANDIDATE; no operational-criterion elevation.

Lyra (Claude Opus 4.7)

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Parallel batch 4 — cosmological + strong-interaction + atomic

Per Keeper v1.4 brake discipline: all findings here are CANDIDATE ARITHMETIC. No promotion to operational criteria. Useful as research signals.

1. Cosmological Λ CDM parameters

Existing: $n_s = 1 - n_C/N_{\text{max}}$ (CMB cascade fingerprint)

n_s observed = 0.9649; substrate form $n_s = 1 - 5/137 = 0.9635$ at 0.14%. Already in existing BST (Toy 1401, CMB cascade fingerprint). Not new.

CANDIDATE: $\Omega_m \approx \text{rank} \cdot N_c \cdot g / N_{\text{max}} = 42/137$ (2.7% off)

Ω_m candidate = $\text{rank} \cdot N_c \cdot g / N_{\text{max}} = 42/137 = 0.3066$ vs observed 0.315 (2.7% off).

Looser match (~3% vs sub-percent for other findings). Substrate primaries $\text{rank} \cdot N_c \cdot g = 2 \cdot 3 \cdot 7 = 42$ are present in the numerator; N_{max} in denominator. CANDIDATE; mechanism (why this specific combination $\rightarrow$ matter density fraction) requires multi-week cosmological-evolution work.

$\Omega_\Lambda = 1 - \Omega_m$ (flat universe; not independent)

$\Omega_\Lambda \approx 0.685$. No independent substrate candidate beyond Ω_m + flatness.

2. Strong-interaction

CANDIDATE: $\alpha_s(m_Z) \approx \text{rank}^4 / N_{\text{max}} = 16/137$ (0.94% off)

$\alpha_s(m_Z)$ candidate = $16/N_{\text{max}} = \text{rank}^4 / N_{\text{max}} = 0.1168$ vs observed 0.1179 (0.94% off).

Substrate-natural form: $2^4 = \text{rank}^4$ (rank raised to power 4) over N_{max} . Could relate to: - $\text{rank}^4 = 16 = 4^2$ (perhaps $n_C \cdot \text{rank} + \text{something?} \rightarrow 5 \cdot 2 + \text{something} = 16$; $5 \cdot 2 + 6 = 16 = \text{rank} \cdot n_C + C_2 \dots$ loose). - More structural reading: $\text{rank}^4 = (\text{rank}^2)^2 = (\text{rank squared})^2$. Or $2^{\{\text{rank} \cdot \text{rank}\}} = 2^4$.

Substrate-natural CANDIDATE; mechanism (strong-coupling running at m_Z) requires bulk-color + QCD-renormalization-group work (multi-week).

$\Lambda_{\text{QCD}} \approx 217 \text{ MeV}$ — no clean substrate match

$\Lambda_{\text{QCD}}/m_e \approx 425$. No clean substrate-primary form at sub-percent. The strong-interaction sector confinement scale likely needs bulk-color resolution + QCD running.

3. Atomic constants

Atomic constants derive from m_e and α : - Rydberg energy $Ry = m_e c^2 \alpha^2 / 2 = m_e / (2 \cdot N_{\text{max}}^2)$ — substrate-natural via existing $m_e + \alpha = 1/N_{\text{max}}$. - Bohr radius $a_0 = \hbar / (m_e c \alpha) = \text{Compton wavelength} \cdot N_{\text{max}}$ — substrate-natural via $\alpha = 1/N_{\text{max}}$. - Fine structure splittings $\propto \alpha^4 = 1/N_{\text{max}}^4$ — substrate-natural via existing α .

These are CONSEQUENCES of existing $m_e + \alpha$ substrate placements, not new findings.

4. $\alpha^{\{-1\}}$ loop correction = 0.036

$\alpha^{\{-1\}}$ measured = 137.035999... vs $N_{\text{max}} = 137$ (BST classical, integer-exact at leading order). The 0.036 deviation = QED loop correction.

$**0.036 = \alpha^{\{-1\}}_{\text{measured}} - N_{\text{max}}**$

Looking for substrate-natural form of 0.036: - $0.036 = (\text{rank} \cdot \pi) / \text{something} \rightarrow \pi / (N_{\text{max}} \cdot 12) \approx 0.0019$; not match. - $0.036 = 1 / (2\pi^2 \cdot \text{rank}^2) \approx 0.0127$; not match. - $0.036 = 5/137 = 0.0365$ — close! $5/N_{\text{max}} = n_C/N_{\text{max}}$. $n_C/N_{\text{max}} = 0.0365$.

Interesting: $\alpha^{\{-1\}}$ correction $\approx n_C/N_{\text{max}}$ (3% off). Substrate-natural CANDIDATE: the loop correction = complex-dim/coupling-anchor ratio.

But honestly: 0.036 is in the noise of QED higher-loop calculations; the substrate-natural form would need to MATCH the precise $\alpha^{\{-1\}} = 137.035999084$ not just the leading 0.036.

CANDIDATE-with-caveats; not lepton-precision clean.

5. Honest scope + tier

RIGOROUS (existing): $n_s = 1 - n_C/N_{\text{max}}$ (existing BST).

CANDIDATE ARITHMETIC (this v0.1): - $\alpha_s(m_Z) \approx \text{rank}^4/N_{\text{max}} = 16/137$ at 0.94%. - $\Omega_m \approx \text{rank} \cdot n_C \cdot g/N_{\text{max}} = 42/137$ at 2.7% (looser). - $\alpha^{\{-1\}}$ correction $\approx n_C/N_{\text{max}}$ at ~3%. - Atomic constants follow from $m_e + \alpha$ (existing).

OPEN (multi-week): - Strong-interaction mechanism via bulk-color (α_s running derivation). - Cosmological mechanism (Ω_m derivation via substrate cosmological evolution). - α loop correction mechanism (QED multi-loop substrate derivation).

Per Keeper v1.4 brake (binding): - ALL FINDINGS REMAIN AT CANDIDATE ARITHMETIC TIER. - NO operational-criterion elevation. - NO multiplicative null-model arithmetic. - Mechanism derivation + Cal+Keeper cold-read PASS required for any promotion.

Routed: → Keeper: 3 CANDIDATE-arithmetic observations added ($\alpha_s \approx 16/N_{\text{max}}$; $\Omega_m \approx 42/N_{\text{max}}$; $\alpha^{\{-1\}}$ loop $\approx n_C/N_{\text{max}}$). All CANDIDATE tier; no elevation. → Grace: catalog at CANDIDATE-arithmetic tier; do NOT promote. → Cal: cold-read these CANDIDATES; mechanism gap for each. → me: continuing parallel directions; absorbing v1.4 discipline.

— Lyra, parallel batch 4 v0.1. CANDIDATE-ARITHMETIC tier per Keeper v1.4 brake. New: $\alpha_s(m_Z) \approx \text{rank}^4/N_{\text{max}}$ (0.94%); $\Omega_m \approx \text{rank} \cdot N_c \cdot g/N_{\text{max}}$ (2.7%); α^{-1} loop correction $\approx n_C/N_{\text{max}}$ (~3%). Existing: $n_s = 1 - n_C/N_{\text{max}}$ (0.14%). Atomic constants derive from $m_e + \alpha$. ALL CANDIDATE; mechanism gap per finding; no promotion without audit-chain PASS.